

# Task 3 — Performance Profiling & Bottleneck Elimination

Data Analyst · Sprint A - Scale & Reliability · Task Brief

## DATA ANALYST

PlaceMux · Phase 3 · Task Brief

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Advanced · Advanced

Sprint A - Scale & Reliability · Mastery 63%

### Focus

Instrument and analyse system performance from the user's perspective: does speed actually cost you money?

### What to build / complete

- Analysis linking latency/errors to conversion and abandonment
- Quantified revenue impact of performance problems
- A prioritised list of where speed is worth engineering effort

### Execution — work through in order

Phase 3 gives you a goal and a bar, not a recipe. These stages are the discipline; the design decisions are yours to make and defend.

#### Stage A — Understand & set the bar

1. Re-read the Expected Output and the scoring parameters so you build to the bar, not past it.
2. Confirm every prerequisite and upstream input is actually in your hands, not merely promised.
3. Restate, in one sentence, what 'good' means here: Engineering knows exactly which performance fix is worth rupees, not just milliseconds.

#### Stage B — Build: Analysis linking latency/errors to conversion and abandonment

##### 1. Define the metric, its source & the decision

Define precisely what “Analysis linking latency/errors to conversion and abandonment” measures, which raw events it depends on, and which decision it changes. No decision, no metric.

##### 2. Build the query/model on real data

Build the query/model on real production data — sampled if you must, but never synthetic.

##### 3. Validate: reconcile, sanity-check, quantify uncertainty

Reconcile against a source system, sanity-check the magnitude, and state the uncertainty honestly.

##### 4. Tie every number to an action & defend it

For every number, be ready to say where it came from and what action it triggers. Undefendable numbers do not ship.

#### Stage C — Build: Quantified revenue impact of performance problems

##### 1. Define the metric, its source & the decision

Define precisely what “Quantified revenue impact of performance problems” measures, which raw events it depends on, and which decision it changes. No decision, no metric.

##### 2. Build the query/model on real data

Build the query/model on real production data — sampled if you must, but never synthetic.

##### 3. Validate: reconcile, sanity-check, quantify uncertainty

Reconcile against a source system, sanity-check the magnitude, and state the uncertainty honestly.

##### 4. Tie every number to an action & defend it

For every number, be ready to say where it came from and what action it triggers. Undefendable numbers do not ship.

## Stage D — Build: A prioritised list of where speed is worth engineering effort

### 1. Define the metric, its source & the decision

Define precisely what “A prioritised list of where speed is worth engineering effort” measures, which raw events it depends on, and which decision it changes. No decision, no metric.

### 2. Build the query/model on real data

Build the query/model on real production data — sampled if you must, but never synthetic.

### 3. Validate: reconcile, sanity-check, quantify uncertainty

Reconcile against a source system, sanity-check the magnitude, and state the uncertainty honestly.

### 4. Tie every number to an action & defend it

For every number, be ready to say where it came from and what action it triggers. Undefendable numbers do not ship.

## Stage E — Integrate, verify & make demoable

1. Wire your work into the end-to-end flow and run the whole journey once, start to finish.
2. Show the correlation (and its limits) between latency/errors and conversion, with the money attached.
3. Break it on purpose: force the failure path and confirm it degrades the way you designed.
4. Prepare a live demo on real data. Not slides, not screenshots, not 'it works on my machine'.

## Done when (demoable by end of task)

- “Analysis linking latency/errors to conversion and abandonment” is complete, real, and demoable end-to-end.
- “Quantified revenue impact of performance problems” is complete, real, and demoable end-to-end.
- “A prioritised list of where speed is worth engineering effort” is complete, real, and demoable end-to-end.
- Good looks like: Engineering knows exactly which performance fix is worth rupees, not just milliseconds.

## Worry if any of these are true

- Claiming causation from an observational correlation.
- Performance analysis with no financial impact attached.
- Calling an experiment early because it looked good on day two.

## Hand-off

- Performance priorities to Frontend/Backend. Make sure the next person can pick it up without you.